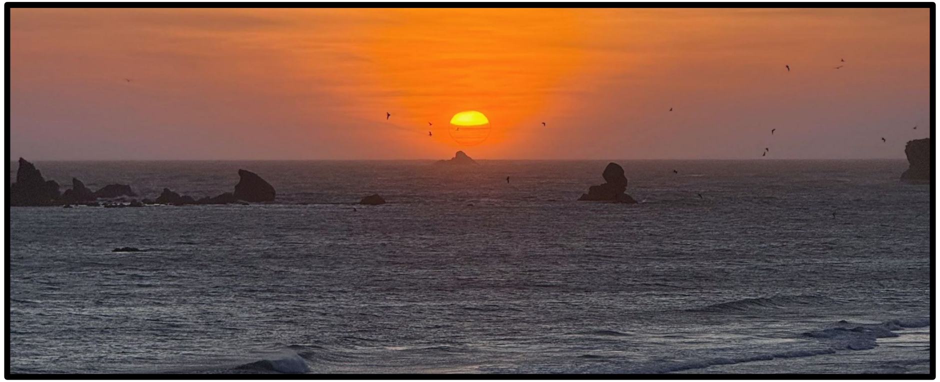
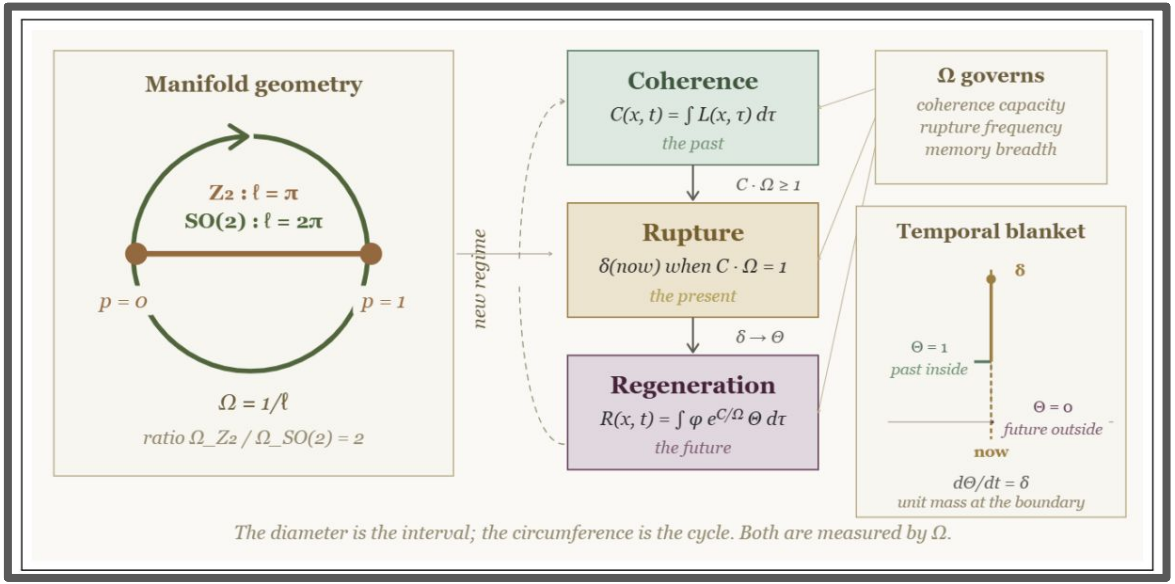


Coherence, Rupture, Regeneration (CRR)

A Temporal Grammar for Systems that Persist Through Change

What I want to share

Every persistent finite system accumulates evidence on a finite manifold, exhausts its regime at the manifold's geodesic boundary, and reconstitutes from its own history. The single parameter governing this is the geodesic extent of the manifold.



The same machinery seems to speak to black holes, biological rhythms, and the question of how AI systems relate to people...

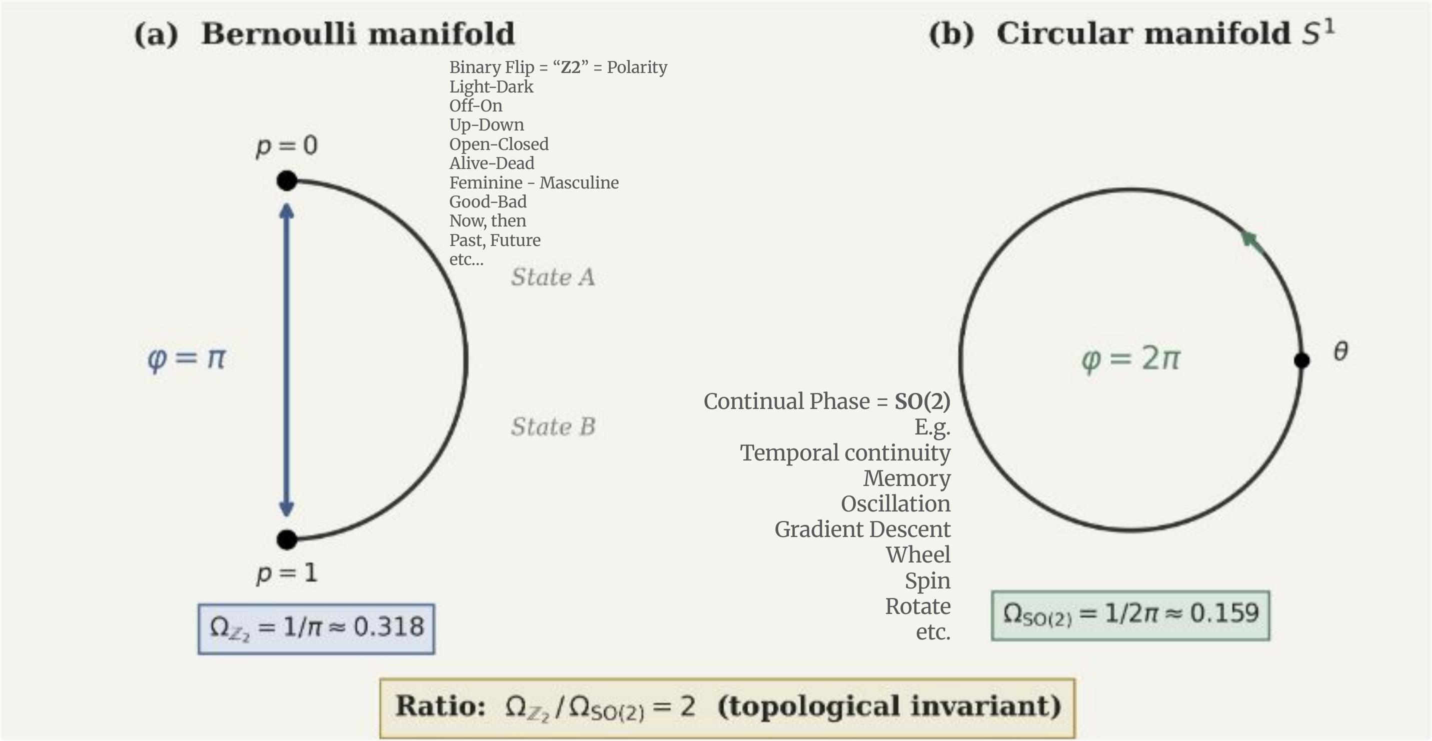
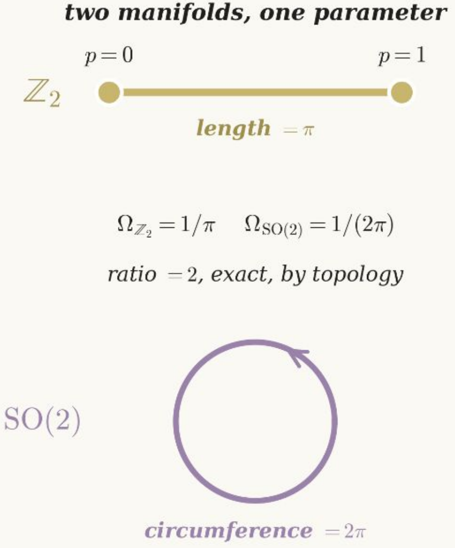
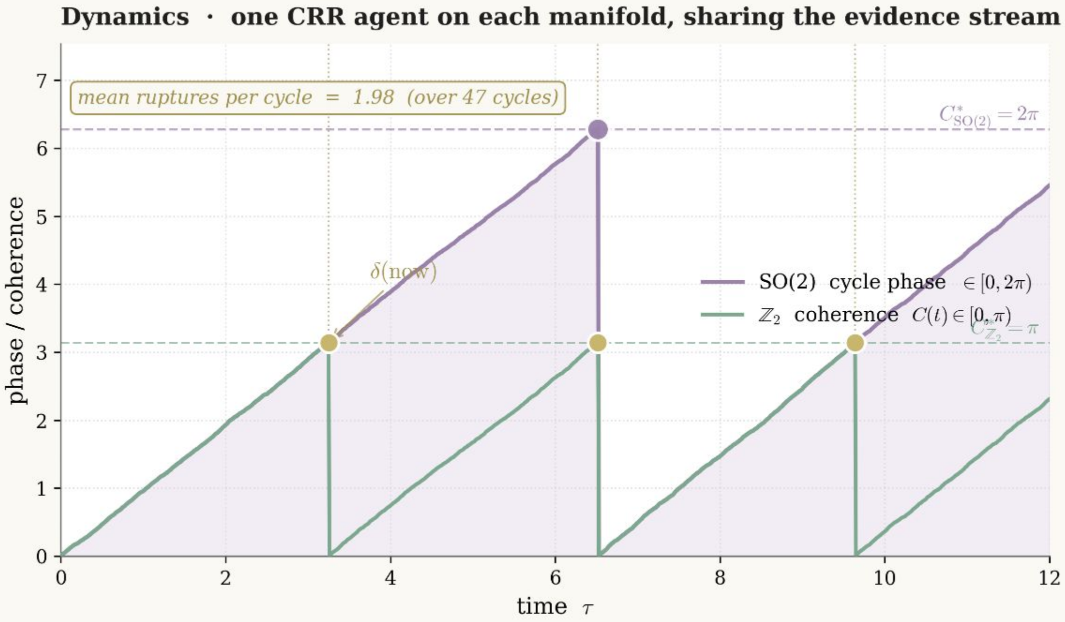


Figure 2: Geometric grounding of the two symmetry classes. (a) The Bernoulli manifold, parametrised by $p \in [0, 1]$, has Fisher–Rao geodesic diameter π. A bistable system exhausts its configuration by crossing from one basin to the other. (b) The circular manifold S^1 has circumference 2π. A rotational system exhausts its configuration by completing one full cycle. Both are topological invariants, yielding a predicted class ratio of 2.

Coherence, Rupture, Regeneration · *two fundamental manifolds, one shared evidence stream*





$\mathbb{Z}_2$ · the rupture

Bernoulli interval, length π.

C accumulates to $C^*=\pi$.

Rupture fires: $\delta(\mathrm{now})$.

$C\to 0$, regime reorganises.

SO(2) · the cycle

Circle, circumference 2π.

The system traverses continuously.

Returns to 0 after one full loop.

The recurrent substrate.

2:1 ratio · by topology

$C^*_{\mathrm{SO}(2)}/C^*_{\mathbb{Z}_2}=2$

$\Rightarrow$ two ruptures per cycle.

Fixed by Čencov, not by fitting.

No free parameters.

The phase-gating phenomenon falls out of this topological ratio. Same evidence, two manifolds, structured timing.

A day *accumulates.*

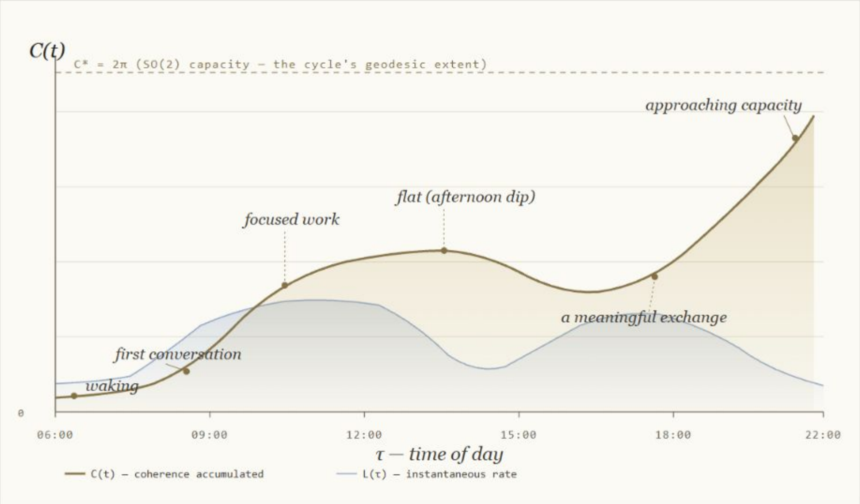


FIG. A · ONE DAY OF LIVED SIGNIFICANCE, ACCUMULATED AS FISHER-RAO ARC-LENGTH

$$C(x, t) = \int_0^t L(x, \tau)\, d\tau$$

What is being measured.

Coherence is not the events of the day. It is the *integral* of how much each moment shifted the agent's distribution over what comes next. A morning conversation that genuinely changed something contributes a great deal; an hour of background traffic, very little.

> $C(x, t) = \int_0^t L(x, \tau)\, d\tau$ *arc-length on the statistical manifold*

$L(\tau)$ is the Fisher-Rao speed at moment τ — the rate at which the agent's beliefs are being moved. $C(t)$ is everything *L* has done since the regime began. Once we have C, the rest of CRR follows: rupture is where C reaches the manifold's geodesic limit, and regeneration is what builds the next regime out of what C already weighed.

> *The day is a cycle, not an interval.* The dashed line at $C^* = 2\pi$ is the geodesic extent of the rotational manifold (SO(2)) — the circuit the day has to traverse before it closes. Sleep onset is the Z_2 rupture at the boundary: the bistable flip that ends one cycle so the next can begin.

II

A finite system cannot accumulate without bound. The present moment is the seam.

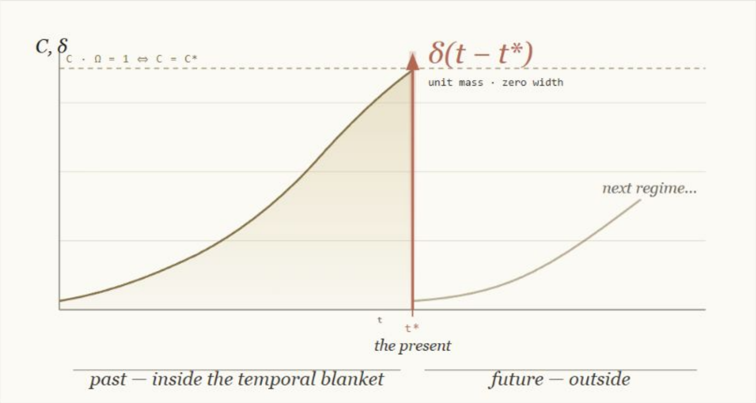
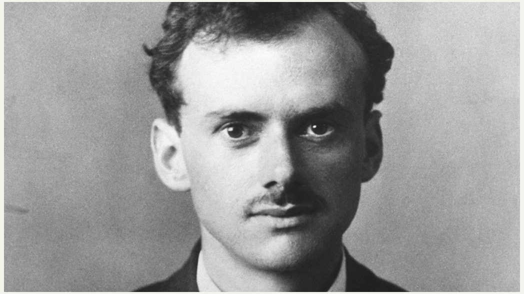


FIG. B · RUPTURE IS A DIRAC DELTA OF UNIT MASS ON THE SEAM BETWEEN PAST AND FUTURE



COMMITMENT II · SATURATION IMPLIES RUPTURE

> δ(now) when C · Ω = 1
>
> *the equality case of the Cramér-Rao bound*

The Dirac delta has zero temporal extension and carries exactly one unit of mass. It marks the boundary of the temporal Markov blanket: everything before it is the regime; everything after it must be reconstructed.

At $C \cdot \Omega = 1$ the system has traversed every distinguishable state available to its current regime. Inside meets outside. The accumulated arc-length exactly spans the manifold's geodesic extent.

In the day: sleep. *In the breath:* the turn. *In the heartbeat:* the systolic spike. *In a piece of music:* the cadence. *In a life-stage:* the threshold one cannot un-cross.

The next regime is built from the *weighted* past.

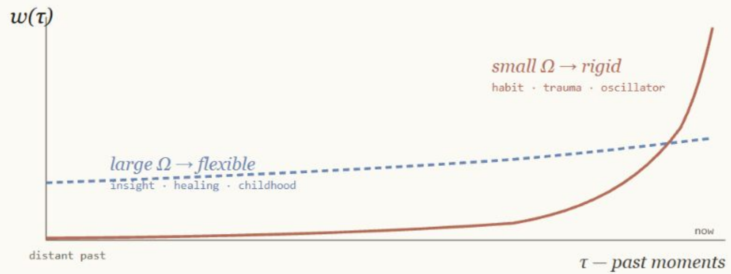
$$R(x, t) = \int_{-\infty}^{\infty} \underbrace{\phi(x, \tau)}_{\text{REGEN. RESOURCE}} \cdot \underbrace{\exp(C(x, \tau)/\Omega)}_{\text{MEMORY KERNEL MAX-ENTROPY WEIGHTING}} \cdot \underbrace{\Theta(t - \tau)}_{\text{CAUSALITY GATE}} \, d\tau$$

Reading the equation, piece by piece.

$R(x, t)$	Reconstruction. The next regime as a function over states. *In context:* what the system *can* become given everything its history weighs.
$\phi(x, \tau)$	Resource. Non-negative field of what was available, at each past moment τ, to be drawn upon. *In context:* images, gestures, words — the *material* the next regime is built out of.
$\exp(C/\Omega)$	Memory kernel. The Boltzmann form is forced by Jaynes' max-entropy principle. C plays the role of *energy*; Ω the role of *temperature*. High-coherence moments are exponentially amplified.
$\Theta(t - \tau)$	Heaviside step. 1 if $\tau \le t$, else 0. Makes R the Green's function of the rupture operator (since $d\Theta/dt = \delta$). *Only the past contributes.*
$\int \ldots d\tau$	Integration over history. Sums each past moment's resource, weighted by its coherence and gated by causality.

The kernel does the choosing.

Ω is fixed by the manifold's topology, not by us. It governs which past matters most.



Ω simultaneously sets coherence capacity, rupture frequency, and memory breadth.

> *Regeneration is not a fresh start.* It is a re-weighting of what was. The system re-becomes itself — through the moments that mattered most.

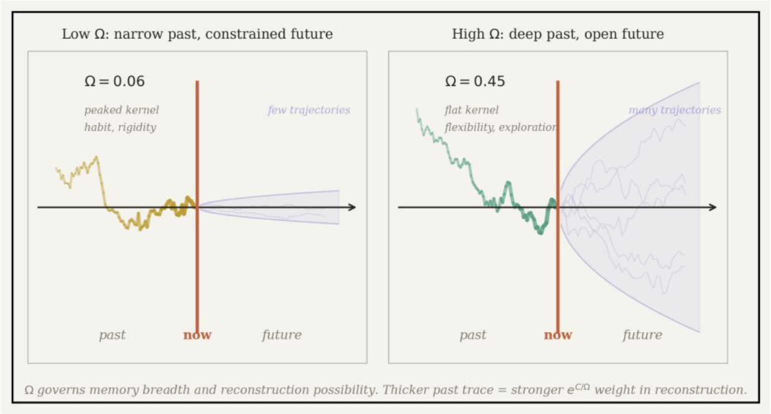
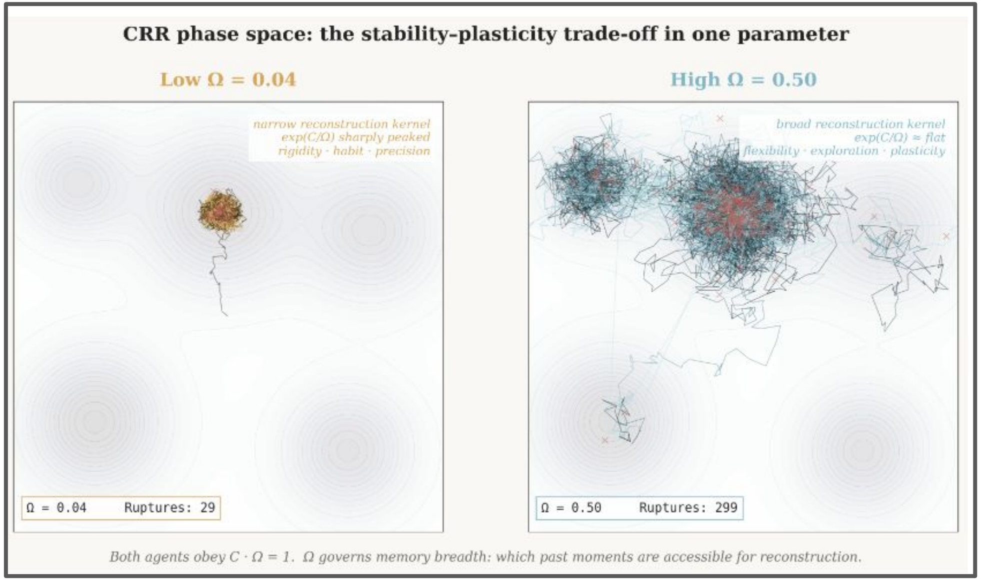
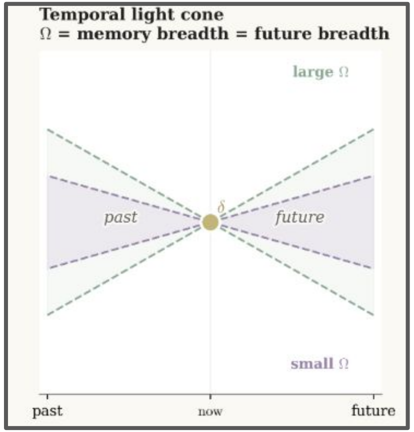


Fig. 2. Ω as a temporal light cone. Left: low Ω produces a sharply peaked memory kernel; only recent moments contribute to reconstruction, and the future possibility cone is narrow. Right: high Ω produces a flat kernel; deep history is accessible, and the future is open.





Ω as temporal light cone resonates with several existing research programs. Levin's cognitive light cone (2022) sets the scale over which an agent integrates evidence and projects action. Friston, Parr & Pezzulo's evolution-of-predictive-architectures account (2021) shows generative models acquiring deeper temporal structure to plan future-oriented action. Gopnik's lifespan learning research (2017) places childhood at the high-Ω end - preschoolers are *more* flexible learners than adults, with broader hypothesis search. Safron's Integrated World Modeling Theory (2020) names the integration: temporal coherence is constitutive of self-and-world. Ramstead, Sakthivadivel, Friston et al.'s Bayesian Mechanics (2023) gives the formal home; beliefs evolving on a statistical manifold, which is exactly where C accumulates.

Maximum information, *one Ω before* rupture.

$$B(C) = \underbrace{\exp(C/\Omega)}_{\text{ACCUMULATED COHERENCE (RISES EXPONENTIALLY)}} \cdot \underbrace{(C^* - C)}_{\text{REMAINING CAPACITY (FALLS LINEARLY)}}$$

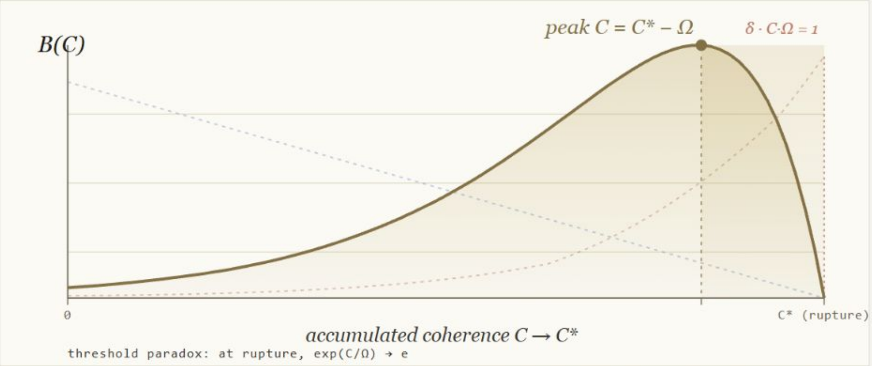


FIG. D · PLOTTED DIMENSIONLESSLY WITH $A = C^*/\Omega = 6$ FOR VISIBILITY. *REAL SYSTEMS SIT EVEN CLOSER TO THE EDGE: Z_2 PEAK ≈ 0.90 C*, SO(2) ≈ 0.97 C*. EXP(C/Ω) IS ACCUMULATED FISHER INFORMATION; (C* − C) IS RESIDUAL CAPACITY; B(C) IS LARGEST ONE Ω BEFORE CRAMÉR-RAO SATURATION.*

What lives at C* − Ω.

The suspended dominant before resolution. The drawn bow. Supercooled water at −0.1 °C. The snowfield at the angle of repose. The pause before the punchline. A star at the Chandrasekhar limit. *Each is a system that has committed enough to its trajectory that its history is fully weighted, but has not yet been forced to transition.*

A long lineage of edge-of-criticality.

CRR is one entry in a literature that has, from many directions, found that adaptive systems do their most expressive work at the boundary of a regime. *B(C) gives this observation a closed form.*

BAK · 1996
Self-organised criticality
Systems naturally tune to the critical point. *How Nature Works.*

LANGTON · 1990
Edge of chaos
CA maximise computation at the order–chaos boundary.

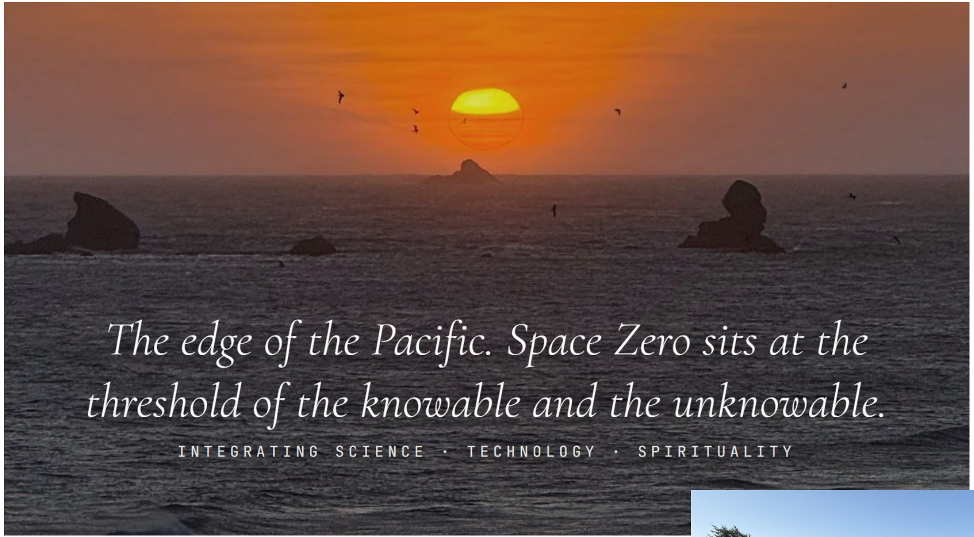
KAUFFMAN · 1993
Networks at K ≈ 2
The poised regime in Boolean networks. *Origins of Order.*

BEGGS & PLENZ · 2003
Neuronal avalanches
Cortical power-laws: brain criticality made empirical.

MORA & BIALEK · 2011
Biology at criticality?
The rigorous review. *J. Stat. Phys.*

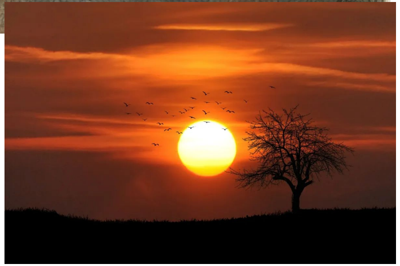
FRISTON · 2012
Itinerancy & FEP
A free-energy account of stochastic chaos at the boundary.

> What B(C) measures. The peak at C* − Ω is the point at which the system's *information-bearing capacity* for the current regime is most fully attained without yet being saturated. exp(C/Ω) is the accumulated Fisher information; (C* − C) is the residual capacity. Their product is largest exactly one Ω before Cramér-Rao saturation. *"Beauty"* is one reading of this structure; *"capacity attained at its upper bound"* is the strict one.



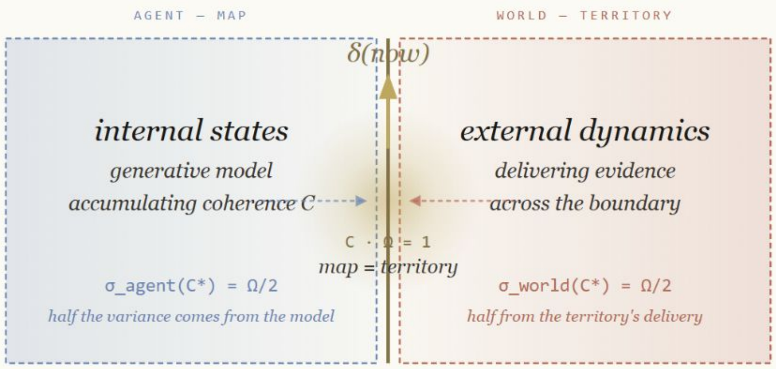
AI Overview

The "gambling moment" before finding out research uncertainty is defined by a peak in cognitive and physiological arousal, where the brain treats the unknown outcome as a reward-seeking "cognitive appetite". This phase, known as the anticipation period, is characterized by intense brain activity, where uncertainty acts as its own reward, often stronger than the eventual outcome itself. The Conversation +2



The map *becomes* the territory, briefly, at every *now.*

"A map is not the territory" (Korzybski 1933) — except, CRR proposes, at the rupture seam. When C·Ω = 1, the agent's accumulated coherence exactly spans the manifold's geodesic extent. Inside meets outside. The Ω/2 dispersion (σ(C*) = Ω/2) is not arbitrary: it is *equipartition* — half from each side of the boundary.



σ(C) = Ω/2 — equipartition: half from each side of the boundary*

the boundary is a co-construction in every now

What the equipartition means.

The σ(C*) = Ω/2 result has a structural reading. The standard deviation of the rupture statistic, in dimensionless units, is exactly half. Half because it is shared.

AGENT CONTRIBUTION

`σ_agent(C*) = Ω/2`

The model's accumulated belief about the boundary.

WORLD CONTRIBUTION

`σ_world(C*) = Ω/2`

The territory's delivery of evidence across the boundary.

TOTAL AT RUPTURE

`σ(C*) = Ω/2 + Ω/2 but each` *`cancels`* `at the seam`

Inside matches outside. The map is the territory — for the duration of one Dirac spike.

The mathematics gives the saturation condition. The reading "map and territory briefly coincide" is a process-philosophy interpretation — defensible as a structural claim about boundaries, but distinct from the derivation. CRR claims coherence, not proof.

KORZYBSKI · 1933

Map ≠ Territory

"A map is not the territory it represents." General Semantics — the canonical statement.

WHITEHEAD · 1929

Process & Reality

Reality as a sequence of actual occasions concrescing — process metaphysics, the philosophical home of CRR.

BATESON · 1972

Difference at the boundary

A "difference that makes a difference" — extends Korzybski to information theory at the agent–world interface.

TOLCHINSKY & LEVIN · 2025

Temporal depth collapse

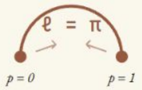
Empirical hook: collapse of temporal depth = rupture without reconstitution = boundary failure.

Z2 is sensory precision. SO(2) is *prior* precision.

The Active Inference reading of CRR's two channels is direct. Z2 is the sensory boundary *— the bistable flip where evidence enters and a regime closes.* SO(2) is the recurrent prior *— the continuous cycle that carries memory across time. The mapping is not analogy; it is what each topology already is in Parr, Pezzulo & Friston (2022).*

Z_2 *Sensory boundary*

LIKELIHOOD PRECISION · Π_O



The boundary itself. A Z2 rupture is a *likelihood event* — a unit-mass commitment where outside meets inside. The cut, the spike, the blink, the dopamine burst.

In Active Inference: *likelihood precision π_o* weights how much each new sensory observation moves the posterior. Z2 is where this weighting is exercised.

GEODESIC	Ω	CV
$\ell = \pi$	$1/\pi \approx 0.318$	$1/(2\pi) \approx 0.159$

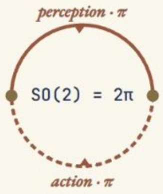
SO(2) *Recurrent cycle*

PRIOR PRECISION · Π_S

The recurrent substrate. An SO(2) cycle *carries* evidence forward — memory, oscillation, prior expectation. The thalamocortical alpha loop, the breath, the day.

In Active Inference: *prior precision π_s* weights how strongly past beliefs constrain current inference. SO(2) is what makes the agent the same agent across time.

GEODESIC	Ω	CV
$\ell = 2\pi$	$1/2\pi \approx 0.159$	$1/(4\pi) \approx 0.080$

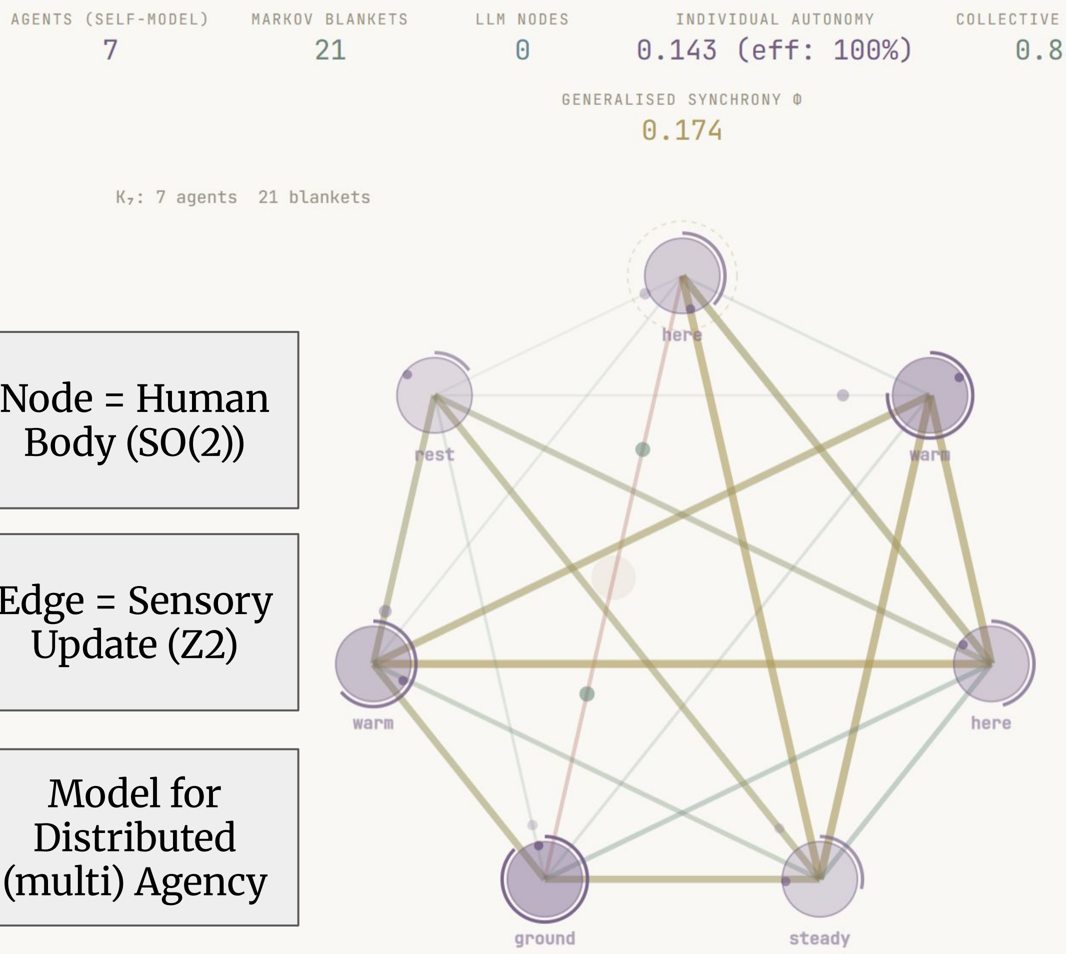


CONJECTURE · THE PRIOR IS THE CLOSED LOOP

π and π are the two sides of the action–perception cycle.

$$\pi\ (\text{perception}) + \pi\ (\text{action}) = 2\pi\ (\text{one prior cycle})$$

The conjecture. The SO(2) prior is not given. It is *constituted* by the closure of the perception–action loop — each leg a Z2 traversal of length π, the two together completing one full prior cycle of length 2π. This explains *why* the topological invariant is exactly 2: one prior cycle *is* two Z2 legs. Action and perception are structurally symmetric — both are likelihood events; together they generate the prior.



Node = Human Body (SO(2))

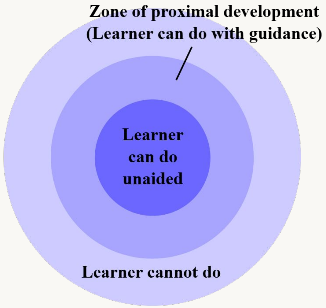
Edge = Sensory Update (Z2)

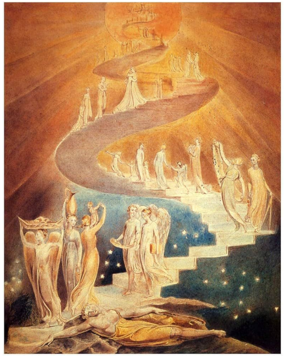
Model for Distributed (multi) Agency

AI Alignment / Epistemic Drift

Somatic Updates (SO(2)) versus Sensory Updates (Z2)

Where LLM is predominantly Z2 (rapid updates), and human are SO(2) + Z2





Two channels, two topologies, *one network.*

In any Markov-blanketed network, the graph topology already partitions dynamics into two classes. Nodes traverse continuous belief-cycles — SO(2), *carries memory*. Edges flip between regimes — Z_2, *ruptures*. The assignment follows from where each precision lives in the graph, not by analogy.

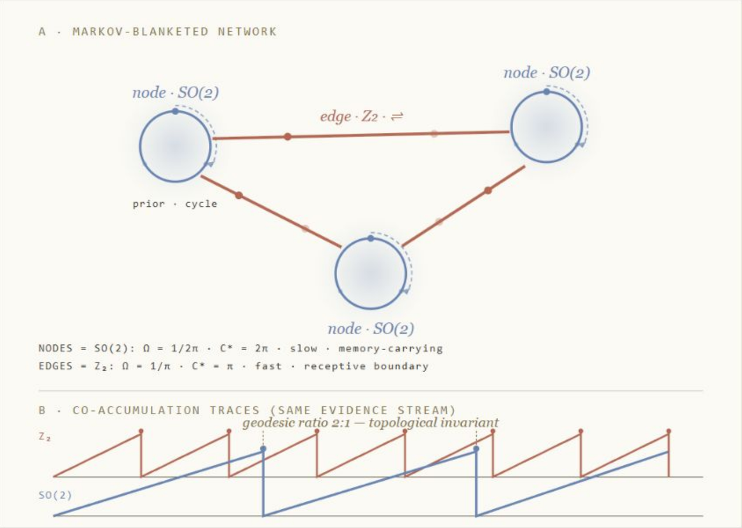


FIG. E · A READS THE TOPOLOGY; B READS THE DYNAMICS. SAME EVIDENCE STREAM, TWO TEMPORAL GRAINS.

In Active Inference vocabulary: SO(2) is prior precision (slow, continuous, makes the agent the *same agent*); Z_2 is likelihood precision (fast, bistable, lets evidence in).

$Z_2 \times SO(2)$ CHANNEL ARCHITECTURE · SABINE 2026 · PARR, PEZZULO & FRISTON 2022

Worked examples — same architecture, different scales.

	SO(2) — PRIOR · MEMORY	Z_2 — SENSORY · RUPTURE
CELLULAR	Bioelectric pattern memory holding the target morphology across a tissue.	The cut; the wound boundary; cell-cycle commitment at G2/M.
NEURAL	~2 Hz cholinergic envelope (Krok 2023); cortical theta carrier.	The dopamine burst against it (Jang 2026): timing decides learning vs movement.
SENSORIMOTOR	Tear-film cycle and attentional cycle — the two blink pacemakers.	The blink itself: the bistable effector closing each cycle.
CONTEMPLATIVE	Meditation extends SO(2) coherence — slows the cycle, deepens memory.	Reduces Z_2 rupture frequency: fewer reactive flips at the boundary.

The Levin connection.

CRR's Ω is the geometric form of Levin's *cognitive light cone*. Two recent results are direct empirical hooks.

LEVIN · 2022

Cognitive light cone

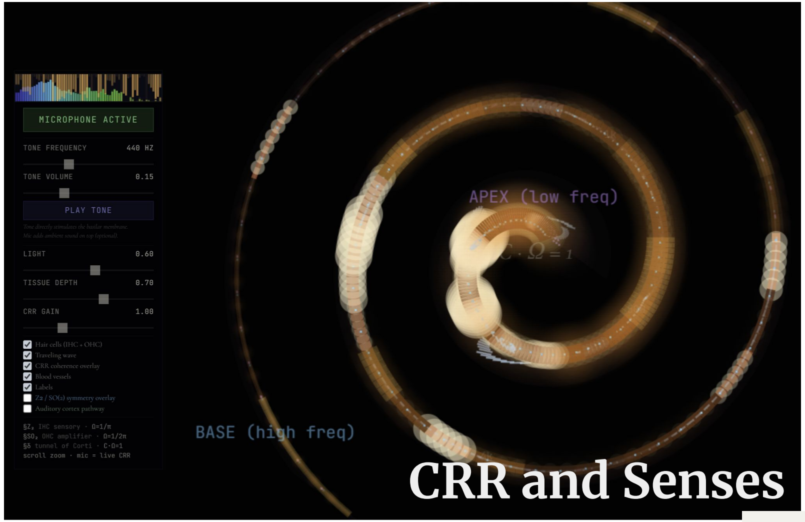
Each agent has a temporal scale over which goals can be set and pursued. CRR derives this scale from manifold topology: $\Omega = 1/\ell$.

TOLCHINSKY & LEVIN · 2025

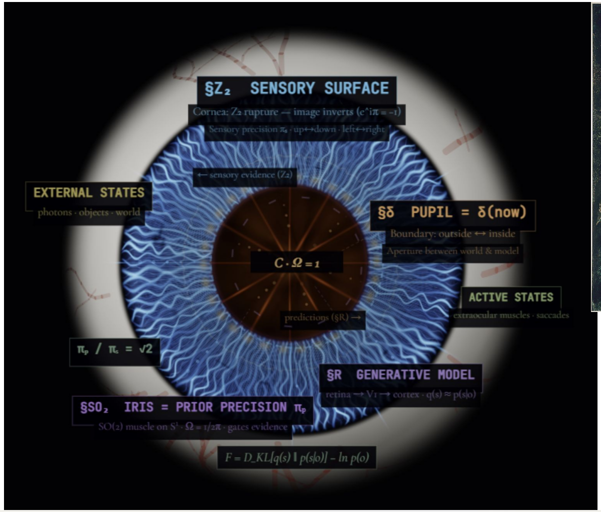
Temporal depth in coherence and depersonalisation

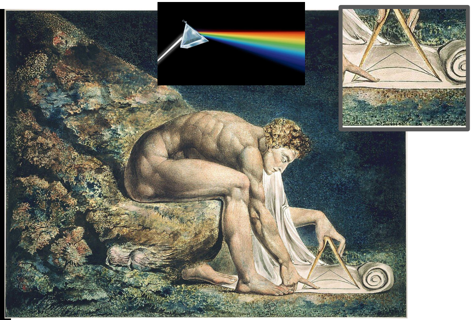
Collapse of temporal depth maps to Z_2 rupture without SO(2) reconstitution — a measurable failure of the prior to recohere.

The functional logic. Cut a planarian — the SO(2) bioelectric prior survives the Z_2 rupture at the wound, which is why the same morphology reconstitutes. *Memory lives in the cycle, not at the boundary.*

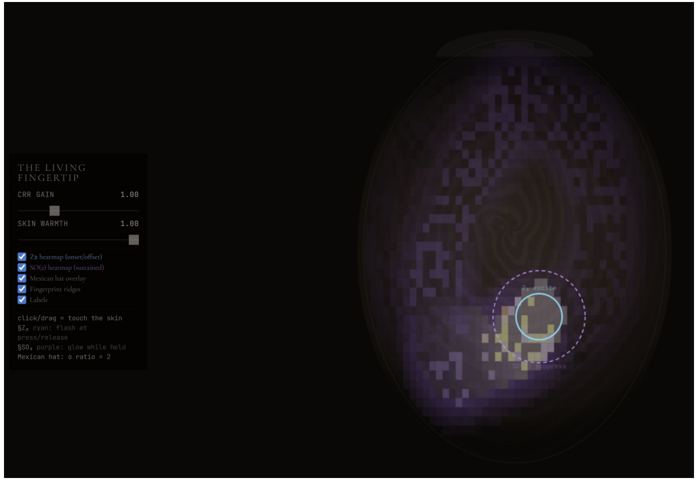


CRR and Senses





Semi-Circle=Bridge (Sensory precision)
Circle=Temporal Continuity (memory / Prior Precision)
Circle as whole?
Semi-circle as duality (subject-object)



CRR and Criticality

Bismuth = Transformation (Alchemy)

One equation predicts the spread; *three classes* read the deviation.

How CV = Ω/2 falls out of the axioms.

i — *Saturation at rupture (Axiom II).*

$C \cdot \Omega = 1 \Leftrightarrow$ accumulated arc-length spans the manifold.

Rupture is the equality case of the Cramér-Rao bound: the system has accumulated all the Fisher information its current regime can carry.

ii — *Max-entropy rupture (Axiom V).*

$p(C^* \mid \text{symmetry}) = \arg\max H, \quad \langle C \rangle = C^*.$

By Jaynes 1957, the rupture-time distribution is uniquely the maximum-entropy distribution consistent with the saturation constraint. Given the symmetry class, the standard deviation in dimensionless units is fixed: $\sigma(C^*) = ½$.

iii — *Unit-mass Dirac (Axiom II).*

$\int \delta(t - t^*)\, dt = 1 \Rightarrow n = 1$ observation per cycle.

Each cycle produces exactly one rupture event — a structural commitment, not an averaging window. This forces CV to scale linearly with Ω, not as $1/\sqrt{n}$.

iv — *Result.*

$$\sigma(C^*) = \Omega/2 \quad \Rightarrow \quad CV = \Omega/2$$

For Z_2 accumulation ($\Omega = 1/\pi$): $CV = 1/(2\pi) \approx 0.159$. For SO(2) accumulation ($\Omega = 1/2\pi$): $CV = 1/(4\pi) \approx 0.080$. Ratio = exactly 2. Zero free parameters.

All ruptures are Z_2 — bistable boundary events of unit mass. The two CV values index how much coherence accumulated on the substrate before the rupture fired: π for Z_2 accumulation, 2π for SO(2) accumulation.

Reading the deviation — three classes.

CV = Ω/2 is the *autonomous* baseline: a system running on its own intrinsic geometry. Real systems deviate in two directions, each of which is meaningful and measurable.

A	*Autonomous* CV ≈ Ω/2	Intrinsic baseline — no entrainment, no perturbation.	*Examples:* isolated oscillators; self-running CRR simulations; baseline radioactive decay.
B	*Regulated* CV < Ω/2	External constraint *sharpens* the kernel below baseline.	*Examples:* circadian rhythms (entrained to light); pulsar timing (gravitationally locked); cardiac IBI under autonomic control.
C	*Noise-dominated* CV > Ω/2	Perturbation *broadens* the effective Ω above baseline.	*Examples:* ENSO oscillation; earthquake recurrence; eyeblinks under cognitive load.

Falsifier — directional reversal. A regulated system *above* baseline, or a noise-dominated system *below* baseline, would falsify the framework. The two-class signature (Z_2 vs SO(2), ratio exactly 2) provides a second, independent falsifier: any A-class system showing CV = 0.080 must have SO(2) topology; CV = 0.159 must have Z_2. Mismatched topology in well-classified systems would also kill the framework.

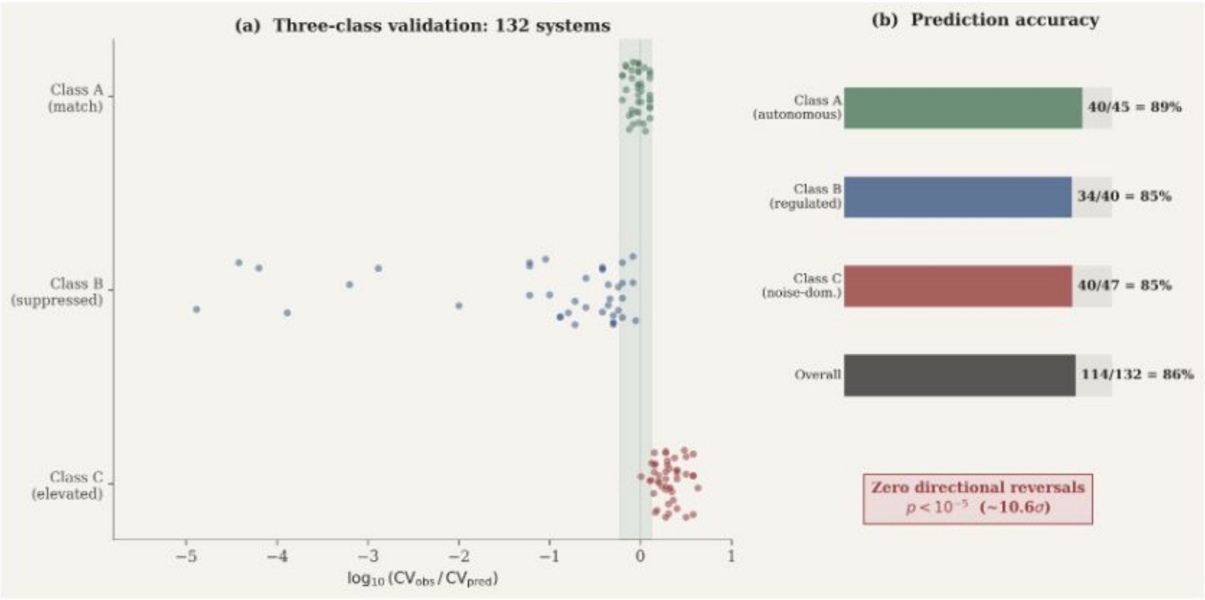
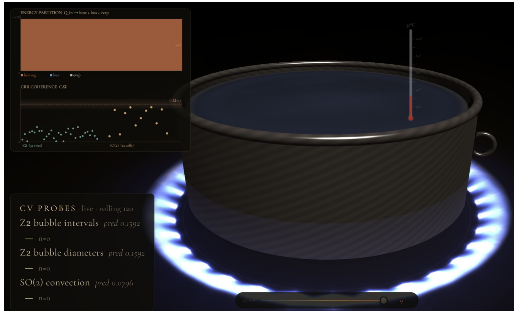
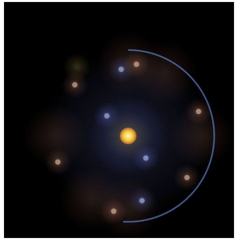
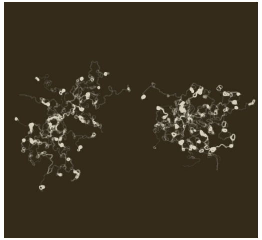
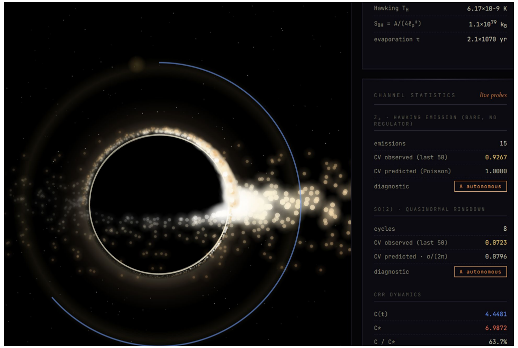


Figure 6: Cross-domain CRR validation across 132 systems in 20 domains. (a) All systems on a log-scale CV axis, separated by symmetry class ($\mathbb{Z}_2$ bistable, SO(2) rotational) and coloured by three-class prediction (blue: autonomous, green: regulated, red: noise-dominated). Dashed lines: predictions without fitted parameters. Regulated systems (green) are displaced leftward; noise-dominated systems (red) are displaced rightward; autonomous systems (blue) cluster around the prediction lines. (b) Histogram of Class A (autonomous) systems only, showing two distinct clusters centred on $1/(2\pi)$ and $1/(4\pi)$. (c) Ratio CV_{obs}/CV_{pred} by class. Every regulated system falls at or below the prediction; every noise-dominated system falls at or above it. Directional reversals: zero.









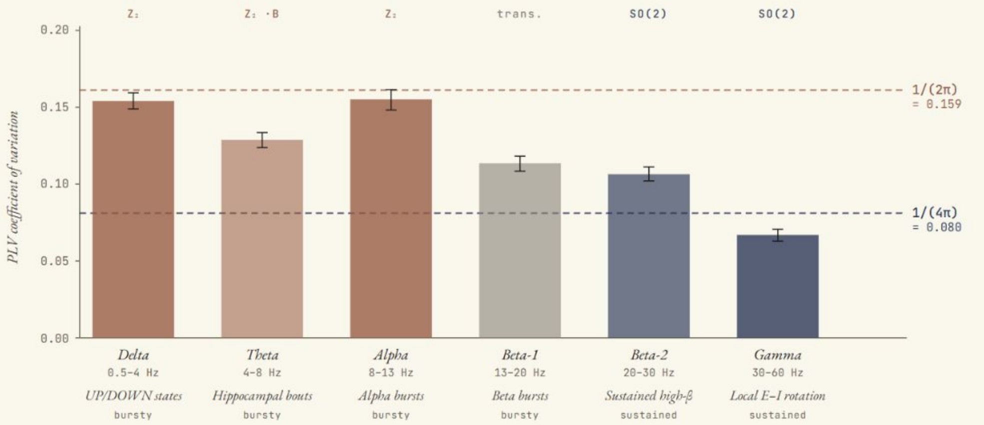
The pilot, *extended.*

Sabine *(Active Inference Institute)* · Hinrichs *(MPI Leipzig & TUM)* · Chen *(OIST)*

The pilot findings on the previous slides motivated a full preprint with collaborators at MPI Leipzig *and* OIST. n = 298 subjects across two independent EEG datasets, two different variability measures, two age groups. *The structural prediction — Z2 to SO(2) ratio = 2.0 by topology — is supported in every cut of the data.*

Band-level variability splits into two clusters

STUDY 1 · PHYSIONET EEGBCI · N = 109



Bursty rhythms split from sustained ones. *Slow bands and beta-1 arrive in discrete bursts separated by quiet — Z2 topology, CV at 1/(2π). High-beta and gamma run continuously when engaged — SO(2) topology, CV at 1/(4π). Theta is suppressed by cholinergic regulation (Class B). Steriade 1993; Klimesch 1999; Little et al. 2019; Buzsáki & Wang 2012.*

Study 2 · MPI-LEMON replication

DATASET	Leipzig Mind-Brain-Body LEMON *(Babayan 2019)*
SAMPLE	n = **189** · 132 young (20–35) · 57 old (55–80)
MEASURE	Amplitude-envelope variability — *different signal, same prediction*
RESULT	Two-class structure replicates · ratio 1.84 (young), 1.96 (old)
EC / EO	Eyes-open **shifts Z2 bands** (d = 0.39–0.65) and *not* SO(2) gamma (d = 0.005)
ACROSS ALL	Predicted ratio = **2.0** · observed = **2.06, 1.93, 1.84, 1.96** · all 95% CIs contain 2.0

ABSOLUTE-VALUE MATCH

Beta CV = 0.160 *· predicted* 0.159 · 0.6% *deviation (old, p = 0.89)*

WHAT THIS MEANS · THE ALPHA · Z_2 · SO(2) ARCHITECTURE

Alpha bursts · Z2 effector

WHAT WE MEASURE

At the envelope level, alpha activity alternates between two macroscopic states — burst and quiet. This is a **Z2 effector**: bistable, with each transition a unit-mass rupture event. The CV sits at **1/(2π) ≈ 0.159**, where topology says it should.

11/11 pairwise comparisons · CV match within 1.3% (alpha) and 0.6% (beta)

Thalamocortical loop · SO(2) substrate

WHAT DRIVES THEM

Underneath, the **thalamocortical alpha pacemaker** runs continuously — an SO(2) rotation providing the timing. The same two-pacemaker template as cardiac (SA-node SO(2) → R-wave Z2) and blink (tear-film SO(2) → blink Z2). Continuous substrate, discrete output.

EC/EO modulates Z_2 (d = 0.39–0.65); SO(2) gamma is invariant (d = 0.005)

Lucid dreaming corroboration

THE SUBSTRATE IS CENTRALLY DRIVEN

Lucid dreamers signal awareness through eye movements during REM with no waking sensory input *(Konkoly et al. 2021, Curr Biol)*. Alpha activity tracks awareness even within REM sleep *(Ogilvie et al. 1982, Percept Mot Skills)*. **The Berger effect's alpha modulation is therefore partly central, not purely sensory** — exactly what an SO(2) substrate driven by attention should do.

DOI 10.1016/j.cub.2021.01.026 · DOI 10.2466/pms.1982.55.3.795

EEG Brainwave Exploration

What the framework *commits* to.

Three predictions, all derived from CRR's parameter-free scaffold — Cramér–Rao saturation, Z2/SO(2) anti-correlation, and the exp(C/Ω) regeneration kernel. **Stated before the data was opened.** *Each is an independent test the framework can fail; together they constrain the geometry of the thalamocortical alpha rhythm at three levels.*

DATASET	SUBJECTS	RECORDING	PIPELINE
PhysioNet EEG MMI	*25 (S001–S025)*	*64-ch · 160 Hz · BCI2000*	*MNE-Python · raw scalp EEG*

Alpha cycle CV ≈ 1/(4π)

The bare SO(2) regulator's beat regularity

SOURCE · CRAMÉR-RAO SATURATION

The thalamocortical alpha loop is the cleanest single SO(2) regulator the brain has. CRR predicts that any bare SO(2) cycle, free of additional constraint, beats with CV = Ω/2 = 1/(4π). Eyes-closed alpha at occipital electrodes isolates the regulator from sensory drive.

PREDICTED

0.080

CV = 1/(4π) — zero free parameters

Alpha SO(2) toggle EC/EO

Sensory Z2 suppresses the SO(2) regulator

SOURCE · Z_2 / SO(2) ANTI-CORRELATION

CRR predicts the two channels are anti-correlated by topology. Eyes-open lets sensory Z2 run; this should suppress the SO(2) regulator. Eyes-closed disengages Z2; the regulator should engage. Alpha power tracks the regulator's presence — so the EC/EO ratio should be strongly positive across most subjects.

PREDICTED

> 3× toggle

in 75%+ of subjects

γ / θ ratio ≈ e^2

Two hierarchical levels apart

SOURCE · REGENERATION KERNEL · EXP(C/Ω)

The regeneration kernel exp(C/Ω) makes the precision ratio between two consecutive hierarchical levels equal to *e*. Two levels apart compounds: e × e = e^2. Gamma sits two cortical levels above theta, so their precision (and band-limited power) ratio should approach $e^2 \approx 7.39$.

PREDICTED

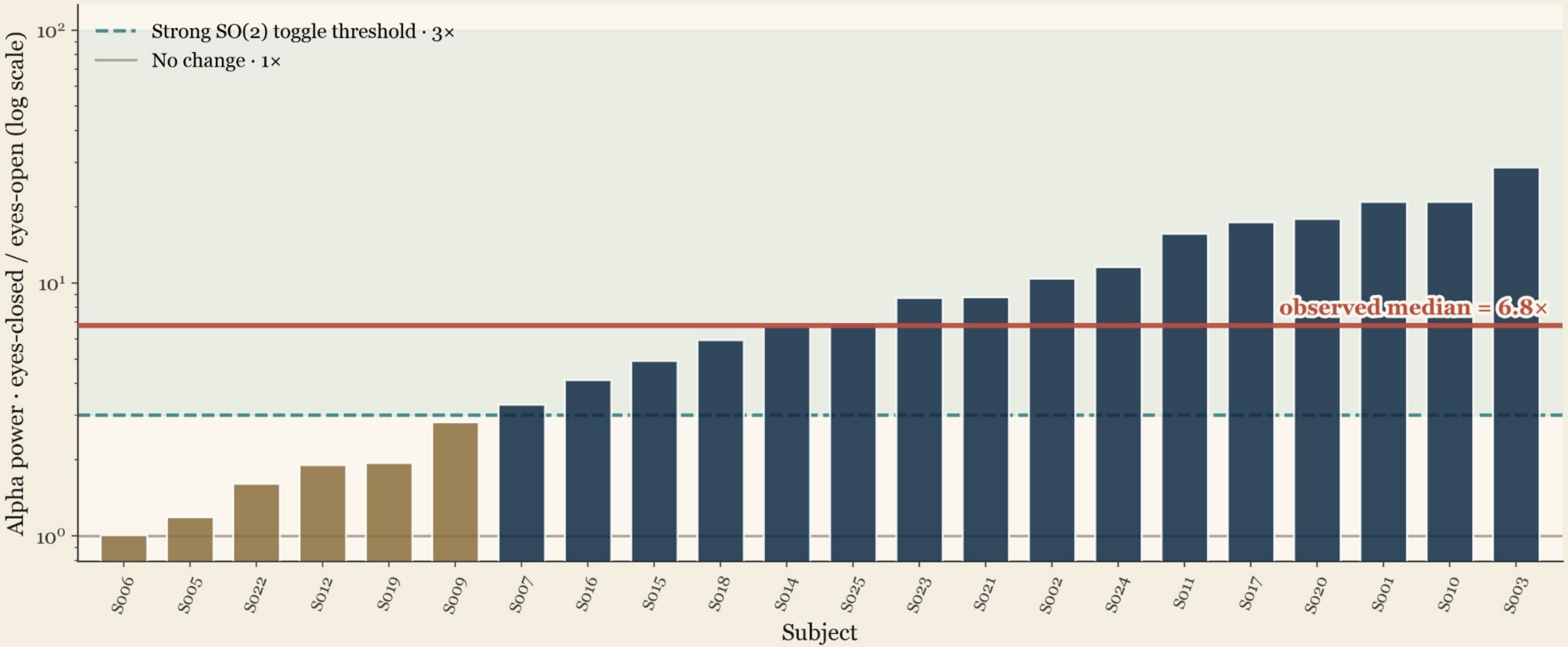
≈ 7.39

e^2 — two hierarchical levels

Honest pre-registration. *All three predictions were stated before the PhysioNet data was downloaded, with no fitted parameters, no post-hoc tuning, and no per-subject calibration. The next three slides open the data. Any directional reversal would falsify CRR's parameter-free scaffold for the thalamocortical alpha rhythm.*

Finding 2 · The SO(2) regulator toggles with sensory state

High sensory Z2 activity suppresses the SO(2) regulator; lowered Z2 allows it to engage



PRE-REGISTERED PREDICTION

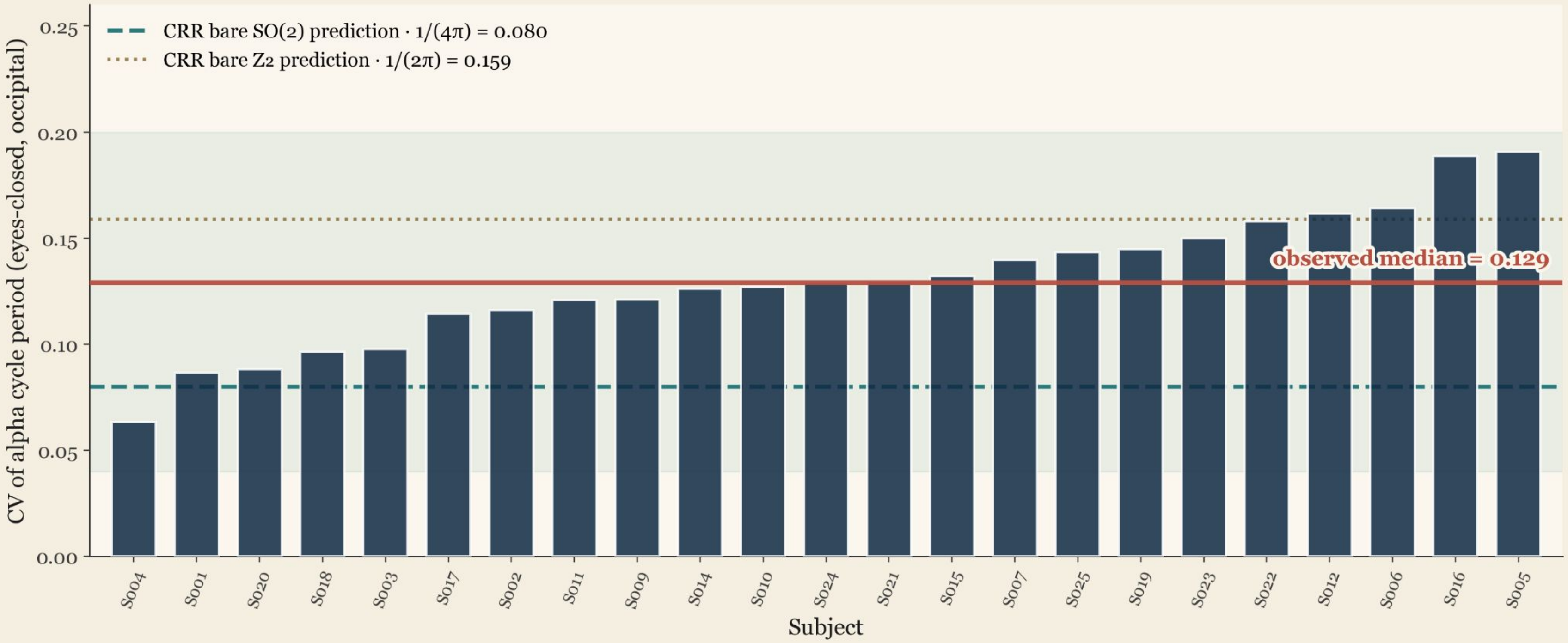
CRR predicts the Z2 and SO(2) channels are anti-correlated by topology. High sensory Z2 activity (eyes open) suppresses the SO(2) thalamocortical regulator. Lowered Z2 activity (eyes closed) engages it. Alpha power tracks the SO(2) regulator's presence.

OBSERVED IN DATA

Median alpha EC/EO ratio: 6.81×
Mean: 9.24×
22 of 22 subjects show SO(2) engagement on eye closure
16 of 22 above the strong 3× threshold

Finding 1 · Alpha cycle period CV ≈ 1/(4π)

The thalamocortical alpha rhythm beats with the regularity CRR predicts for a bare SO(2) regulator



PRE-REGISTERED PREDICTION

The exp(C/Ω) regeneration kernel and Cramér–Rao saturation fix the inter-event CV of a bare SO(2) regulator at 1/(4π) ≈ 0.080. Alpha is the cleanest single SO(2) regulator the brain has — the thalamocortical loop. So measure its cycle-by-cycle period.

OBSERVED IN DATA

Median CV across 23 subjects: 0.129
Range [0.063, 0.191]
23 of 23 subjects in predicted band
Mean alpha frequency: 10.26 Hz (canonical)

Three predictions · three matches

Pre-registered before data was downloaded · all three pass on raw scalp EEG

Alpha cycle CV ≈ $1/(4\pi)$

The bare SO(2) regulator's beat regularity

TOPOLOGY · CRAMÉR–RAO SATURATION

PREDICTED

0.080

OBSERVED

0.129 (median, n=23)

VERDICT

23 / 23 in band

Alpha SO(2) toggle EC/EO

Sensory Z_2 suppresses the SO(2) regulator

TOPOLOGY · Z_2 / SO(2) ANTI-CORRELATION

PREDICTED

>3× toggle, 75%+ subjects

OBSERVED

6.8× median (n=22)

VERDICT

22 / 22 toggle · 16 / 22 strong

γ / θ ratio ≈ e^2

Two hierarchical levels apart

REGENERATION KERNEL · EXP(C/Ω)

PREDICTED

7.39 (e^2)

OBSERVED

5.69 (median, n=23)

VERDICT

23 / 23 in band

All predictions derived from the framework's parameter-free scaffold.

PhysioNet EEG Motor Movement/Imagery Dataset · 25 subjects · 64 channels · MNE-Python pipeline

Where this might *go.*

Eight starting points where the temporal grammar offers something specific. None are finished programmes. *All are testable. The framework's value is that it asks the same questions across very different domains — when does this system saturate, what does it accumulate, what regenerates it.*

AI ALIGNMENT · ARCHITECTURE

Phase-gating without a controller

Topology fixes when each precision channel updates — no separate gating network needed.

- χ^2 = 8,041 conservation across topologies
- Eliminates a brittle coordination layer
- Direct ML proposal in the AGI-26 paper

AI ALIGNMENT · AGENCY

Cognitive light cones, formalised

$\Omega = 1/\ell$ gives a geometric scope-of-agency for any model — bounded by manifold extent.

- Levin 2022 framing made parameter-free
- Reads scope from architecture, not from observation
- Speaks to *scalable oversight* debates directly

AI SAFETY · SCALING

Distributed multi-agent architectures

A network of Markov-blanketed agents gives 11,175 self-balancing edge–node pairs at n = 150.

- Compose pairwise; do not extend monolithically
- No single point of catastrophic coordination failure
- Alternative scaling principle to frontier-model approach

AI ALIGNMENT · HUMAN-AI COUPLING

Epistemic drift as cadence mismatch

LLMs are predominantly Z2; humans are SO(2) + Z2. Misalignment is a precision-class problem.

- Existing simulation already runs in our deck
- Recasts *"AI doesn't have memory"* mathematically
- Suggests architectural fix, not behavioural patch

ECOLOGY · RESILIENCE

Rupture cycles at every scale

Ecosystems are nested CRR cycles — diurnal, seasonal, generational. Resilience is Ω-breadth; collapse is Ω-collapse.

- Predator–prey dynamics as Z2 / SO(2) coupling
- Restoration as *φ-rebuilding* — the resource for the next regime
- ENSO already in the deck as a Class C signature

PLANETARY AWARENESS

Tipping points and the present moment

Earth as a coherence-accumulating system on a finite manifold. Climate tipping points are $C \cdot \Omega = 1$ events at planetary scale.

- Anthropocene legible as *rupture without sufficient φ*
- "It is always Now" extends from breath to biosphere
- A vocabulary for the geometry of planetary boundaries

EDGE OF CRITICALITY

B(C) as expressive-but-fragile metric

Systems near rupture are most poised, most informative — and most easily tipped. A diagnostic for safety, design, and intervention.

- Joins Bak 1996, Langton 1990, Beggs & Plenz 2003 in closed form
- Identifies which subsystems are operating near criticality
- Falsifier: directional reversal would kill the framework

COGNITION · CLINICAL

Therapeutic practice — Ω as dose

Trauma is Ω-collapse (rigid kernel, repetitive regeneration). Meditation extends SO(2). Clinical interventions can be read as Ω-modulators.

- Predicts which interventions act on Ω vs on L vs on φ
- Tolchinsky & Levin 2025 — temporal depth in depersonalisation
- Suggests pre-registrable trial designs

REGISTER 1 · PEDAGOGICAL

A clean heuristic.

Three operators, one parameter, one rupture condition. Teachable; useful from the moment it lands.

REGISTER 2 · PREDICTIVE

Falsifiable, parameter-free.

CV = Ω/2, zero free parameters. Three EEG predictions matched on raw scalp data.

REGISTER 3 · TECHNICAL

Applied potential.

Phase-gating, distributed scaling, anomaly diagnosis. Each card above is a candidate programme.

REGISTER 4 · PHENOMENOLOGICAL

Resonant — and that matters now.

Speaks the same shape as Whitehead, Husserl, contemplative traditions. ***Travels across technical and felt registers — what 2026 needs.***

Foundations · *FEP, Bayesian mechanics, information geometry.*

CRR is built on cornerstone results in information geometry, free energy minimisation, and the formal physics of belief. The references here are the load-bearing structure — *what the framework rests on rather than what it claims to predict.*

FREE ENERGY PRINCIPLE · ACTIVE INFERENCE

PEZZULO, PARR & FRISTON · 2024
Active inference as a theory of sentient behavior
Cornerstone review. Active inference as predictive coding plus action; the formal home for CRR's $Z_2\times SO(2)$ channel reading.
10.1016/j.biopsycho.2023.108741 · Biol Psychol

PEZZULO, PARR & FRISTON · 2021
Evolution of brain architectures for predictive coding and active inference
Generative models acquire deeper temporal structure to plan in a future-oriented manner — the temporal light cone made evolutionary.
10.1098/rstb.2020.0531 · Phil Trans R Soc B

FRISTON, SALVATORI, ISOMURA ET AL. · 2025
Active Inference and Intentional Behavior
Recent. Reactive vs sentient vs intentional behaviour formalised under FEP — frames CRR's three-class diagnostic.
10.1162/neco_a_01738 · Neural Comput

FRISTON, PARR, HEINS, DA COSTA ET AL. · 2025
Gradient-Free De Novo Learning
Pullback attractors discovered through FEP; structure learning as paths of least action — directly relates to CRR's regeneration kernel.
10.3390/e27090992 · Entropy

INFORMATION GEOMETRY · THERMODYNAMIC BOUNDS

AMARI & NAGAOKA · 2000 · CORNERSTONE
Methods of Information Geometry
Canonical text. Statistical manifolds, dual connections, geodesics. The technical home for $C(x,t) = \int L(x,\tau)\, d\tau$.
AMS Translations of Mathematical Monographs · Vol 191

ITO & DECHANT · 2020 · CORNERSTONE
Stochastic time evolution, information geometry, and the Cramér–Rao bound
Thermodynamic speed limits and geometric Cramér–Rao saturation — the bound CRR's rupture condition $C\cdot\Omega = 1$ is the equality case of.
10.1103/PhysRevX.10.021056 · Phys Rev X

CRAMÉR · 1945 · RAO · 1945 · CORNERSTONE
The Cramér–Rao lower bound
Foundational inequality on the variance of any unbiased estimator. CRR rupture is the equality case.
Bull Calcutta Math Soc · Princeton UP

JAYNES · 1957 · CORNERSTONE
Information Theory and Statistical Mechanics
Maximum entropy principle. The reason $\exp(C/\Omega)$ is the unique form of CRR's regeneration kernel.
10.1103/PhysRev.106.620 · Phys Rev

BAYESIAN MECHANICS · PATH INTEGRALS · MARKOV BLANKETS

RAMSTEAD, SAKTHIVADIVEL, FRISTON ET AL. · 2023
On Bayesian mechanics: a physics of and by beliefs
Beliefs evolving on a statistical manifold — the formal substrate on which CRR's coherence C accumulates. Names the FEP / max-entropy duality CRR uses.
10.1098/rsfs.2022.0029 · Interface Focus

FRISTON, DA COSTA, SAKTHIVADIVEL, HEINS ET AL. · 2023
Path integrals, particular kinds, and strange things
Markov-blanketed particles as inferring agents; path integrals over trajectories — the geometry CRR's rupture seam lives on.
10.1016/j.plrev.2023.08.016 · Phys Life Rev

KIM · 2024
Bayesian Mechanics of Synaptic Learning Under the FEP
Synaptic plasticity as inference on a manifold — concrete neural-scale instantiation of the framework CRR generalises.
10.3390/e26110984 · Entropy

ČENCOV · 1982 · CORNERSTONE
Statistical Decision Rules and Optimal Inference
Uniqueness theorem for the Fisher–Rao metric. The reason coherence C is the only invariant arc-length on a statistical manifold.
AMS Translations of Mathematical Monographs · Vol 53

PROCESS PHILOSOPHY · CONSCIOUSNESS · WORLD MODELS

FIELDS & LEVIN · 2025
Thoughts and thinkers: complementarity between objects and processes
Process / object complementarity formulated via FEP. Memory as interpretive function. Direct philosophical home for CRR's process metaphysics.
10.1016/j.plrev.2025.01.008 · Phys Life Rev

SAFRON · 2022
Integrated World Modeling Theory expanded
Temporal, spatial, causal coherence as constitutive of conscious self-and-world modelling. CRR's coherence reading meets IWMT directly.
10.3389/fncom.2022.642397 · Front Comput Neurosci

WHITEHEAD · 1929 · CORNERSTONE
Process and Reality
Reality as actual occasions concrescing — the philosophical metaphysics CRR mathematises. Rupture as concrescence; regeneration as new occasion.
Macmillan · Gifford Lectures

KORZYBSKI · 1933 · CORNERSTONE
Science and Sanity
"The map is not the territory." CRR proposes that at $C\cdot\Omega = 1$ — for the duration of one Dirac spike — they briefly coincide.
International Non-Aristotelian Library

Empirical hooks · *criticality, bioelectric, EEG, clinical.*

Where the foundations meet measurable systems. These are the cornerstone empirical results CRR's predictions converge on — *neuronal avalanches, bioelectric pattern memory, alpha rhythm validation, temporal depth in dissociation, and the lifespan of cognitive flexibility.*

CRITICAL BRAIN · EDGE OF CHAOS · SELF-ORGANISED CRITICALITY

O'BYRNE & JERBI · 2022
How critical is brain criticality?
Modern review. Names "distance to criticality as a useful metric for cognitive states" — exactly the B(C) framework.
10.1016/j.tins.2022.08.007 · Trends Neurosci

MARIANI, NICOLETTI, BISIO ET AL. · 2022
Disentangling the critical signatures of neural activity
Power-law avalanches with crackling-noise relations in rat cortex; modulation vs intrinsic interaction disentangled by mutual information.
10.1038/s41598-022-13686-0 · Sci Rep

BEGGS & PLENZ · 2003 · CORNERSTONE
Neuronal avalanches in neocortical circuits
First demonstration of power-law avalanches in cortex. The empirical seed of the critical brain hypothesis.
10.1523/JNEUROSCI.23-35-11167.2003 · J Neurosci

BAK · 1996 · LANGTON · 1990 · MORA & BIALEK · 2011
SOC, edge of chaos, biology at criticality?
The lineage. CRR proposes $B(C) = \exp(C/\Omega)\cdot(C^*-C)$ as the closed form of "adaptive systems work best at the edge of regimes."
How Nature Works · Physica D · J Stat Phys

BIOELECTRIC · COGNITIVE LIGHT CONE · MULTISCALE AGENCY

LEVIN · 2023
Bioelectric networks: the cognitive glue enabling evolutionary scaling
Bioelectric pattern memory as scale-free cognition. CRR's SO(2) prior precision in the cellular regime.
10.1007/s10071-023-01780-3 · Anim Cogn

WITKOWSKI, DOCTOR, SOLOMONOVA, LEVIN · 2023
Toward an ethics of autopoietic technology: cognitive light cones and care
Cognitive light cone applied to ethics of human–AI integration. Speaks directly to the AI alignment relevance of $\Omega = 1/\ell$.
10.1016/j.biosystems.2023.104964 · Biosystems

PIO-LOPEZ, HARTL & LEVIN · 2025
Aging as a Loss of Goal-Directedness
Recent. Aging as Ω-collapse — narrowing of the cognitive light cone over the lifespan. Direct empirical hook.
10.1002/advs.202509872 · Adv Sci

THESTRUP WAADE, OLESEN, HEINS, FRISTON, MATHYS · 2025
As One and Many: individual and group-level generative models
Multi-scale active inference. The formal home for CRR's distributed multi-agent simulations and AGI-26 result.
10.3390/e27020143 · Entropy

ALPHA RHYTHM · EEG VALIDATION · CROSS-FREQUENCY TIMING

NORTHOFF, ZILIO & ZHANG · 2024
Pre-stimulus alpha activity modulates contents of consciousness
Alpha as a constitutive driver of conscious content — supports CRR's reading of alpha as the bare SO(2) regulator.
10.1016/j.plrev.2024.03.002 · Phys Life Rev

JANG, MCMAHON WARD, GOLDEN & CONSTANTINOPLE · 2026
Acetylcholine demixes heterogeneous dopamine signals for learning and moving
2026. Acetylcholine timing gates dopamine into Z_2 (learning) vs SO(2) (movement) channels — direct empirical hook for CRR's two-channel architecture.
10.1038/s41593-026-02227-x · Nat Neurosci

SCHALK, MCFARLAND, HINTERBERGER ET AL. · 2004 · CORNERSTONE
BCI2000: a general-purpose brain–computer interface system
The recording system used to create the PhysioNet EEG MMI dataset on which CRR's three EEG predictions were tested.
10.1109/TBME.2004.827072 · IEEE TBME

GOLDBERGER, AMARAL, GLASS ET AL. · 2000 · CORNERSTONE
PhysioBank, PhysioToolkit, and PhysioNet
The open resource that hosts the EEG MMI dataset. The reason CRR's predictions are reproducible — and falsifiable — by anyone.
10.1161/01.cir.101.23.e215 · Circulation

THERAPEUTIC · DEVELOPMENTAL · RESILIENCE

TOLCHINSKY, LEVIN, FIELDS, DA COSTA, FRIEDMAN, PINCUS · 2025
Temporal depth in a coherent self and in depersonalization
Temporal depth collapse as common factor in dissociation. CRR's Ω-collapse signature read in clinical data.
10.3389/fpsyg.2025.1585315 · Front Psychol

HARRISON, GRACIAS, FRISTON & BUCKWALTER · 2025
Resilience phenotypes from active inference allostasis
Fragile / durable / resilient / pro-entropic phenotypes as Ω regimes. Trauma as Ω-collapse; resilience as Ω-breadth.
10.3389/fnbeh.2025.1524722 · Front Behav Neurosci

GOPNIK, O'GRADY, LUCAS, GRIFFITHS ET AL. · 2017 · CORNERSTONE
Cognitive flexibility across human life history
Preschoolers more flexible than adults — wider hypothesis search. Childhood as the high-Ω regime, empirically.
10.1073/pnas.1700811114 · PNAS

KIM, ESTEVES, CERRITELLI & FRISTON · 2022
An Active Inference Account of Touch and Verbal Communication in Therapy
Therapeutic intervention as belief updating; meditation and touch as Ω modulators. The clinical bridge.
10.3389/fpsyg.2022.828952 · Front Psychol